Variable-Frequency AC System for Detecting Defects in Metal Infrastructure

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I. Abstract

Structural members in all types of infrastructure degrade over time due to internal stress, corrosion, and surface defects that can be difficult to spot. Existing detection methods are limited, either being destructive in nature or prohibitively time-consuming or costly. Other approaches such as visual inspection may not be able to detect damage that is sub-surface or otherwise hidden. This project addresses this issue by proposing an alternative detection mechanism that uses high-frequency alternating current (AC) signals as a non-destructive, inexpensive, rapid, and reliable method for detecting defects in metal components. Ensuring the quality of metal is very important to public safety, cost effective maintenance, and longevity of infrastructure. We have developed a prototypical testing system that exposes a metallic sample to a broad range (4 – 310 kHz) of AC signals and receives the resulting inductive response produced by the metal surface. The system allows for rapid real-time control and analysis. Using the system, signals are sent through samples of metal using a copper transmission coil, and the material response is measured by a pickup coil. These received signals are then compared against controls, and variations from controls are indicative of material damage, such as abrasions, corrosion, or other abnormalities. We will show results on baseline data captured with pristine metal to ensure system stability across a wide range of frequencies as well as subsequent testing conducted on pieces of metal with prescribed abrasions and other defects to evaluate potential differences between the signals. Our results demonstrate the measurable and repeated differences between the control pieces and the defective pieces, showing differences in signal return of over 25% with RMS differences ranging from 2.7% to 14.6%. This research contributes to the field of engineering by making it easier and more efficient to test infrastructure without unnecessary costs or destructive methods. If applied on a large scale, the implications of this work would be safer inspections, extended infrastructure lifespan, and cheaper access to important diagnostic tools.

II. Intro

Preventing structural failure is a fundamental part of the engineering process, as systems are expected to perform reliably over time. However, structural members naturally degrade over time, and this could be accelerated due to unforeseen issues such as internal stress, fatigue, and exposure to the environment [1][2]. Detecting this degradation early is necessary to maintain safety and avoid catastrophic failures [3]. At the same time, there is an ever-growing need for testing methods that are both reliable and cost-effective, which would enable widespread routine inspections to ensure public safety. There are several methods that are currently being used to assess the structural condition of members, but each has notable drawbacks. Broader reviews of non-destructive testing show that commonly used approaches vary wildly in cost, required equipment, material compatibility, and ability to detect surface or subsurface defects [4]. One example is load testing, which involves applying force to a structure and measuring its deformation to determine integrity [5][6]. There are multiple issues with this method, including potential damage being done to the structure, as well as being time-consuming and costly to perform [7]. Another method which is commonly used is visual inspection. Although this is

much more cost-effective and time-saving, it is very limited in its ability to detect internal/subsurface damage [1]. As a result, critical damage may go unnoticed until it is too late. These limitations highlight a significant gap in existing inspection techniques. There is a lack of a method that is both able to accurately detect defects while also being non-destructive, portable, and reliable in defect detection. While numerous non-destructive testing (NDT) techniques are available for defect detection, such as ultrasonic testing and thermography, each method presents inherent limitations, including material restrictions, sensitivity to noise, and the need for experienced operators [4]. These constraints can complicate implementation in practical inspection settings, particularly when consistent and large-scale monitoring is required. Because of the lack of a product that mitigates these concerns, we have constructed a non-destructive testing system that utilizes high frequency AC currents to detect defects within metals. The development of this testing method enables applications in structural health monitoring of metallic systems, including components in bridges, buildings, and aerospace structures [4]. In these settings, early detection of defects such as cracks, corrosion, and material loss is critical for maintaining safety and performance. The approach is particularly relevant for conductive materials where subsurface defects may not be visible, with sensitivity depending on defect size and depth as well as the electromagnetic properties of the material. In this study, the system demonstrated clear detection of large, through-thickness defects, while smaller defects were closer to experimental uncertainty, and the method is most effective for highly conductive, non-ferrous materials such as aluminum. The testing system exposes the surface of the test metal to an AC field and records the response using a pickup coil [8]. Because the depth that the signal can penetrate into a metal (skin depth) depends on the frequency, a wide frequency range of 4 kHz to 310 kHz, chosen based on hardware limitations, is used to reach from 1.08 mm to 0.118 mm. This approach is a rapid, reliable, and inexpensive alternative to existing methods, addressing the ever-growing need to contribute to safer and more efficient inspections of engineering systems.

III. Theory

Induced Magnetic Field and Coil Response

The testing system that was developed operates on the principles of electromagnetic induction. Electromagnetic induction describes how an AC magnetic field produces an electric response in a conductor [9]. A conductor carrying a current—such as our copper transmission coil—generates a time-varying magnetic field. For a coil with *N* turns, the strength of the field is proportional to the current *I.* This magnetic field extends into the surrounding space and penetrates nearby metal such as our test pieces. As the magnetic field varies in time, eddy currents are induced in the test material. These eddy currents generate their own magnetic field which opposes the field generated by the transmission coil, changing the overall flux in the area. The pickup coil then detects this interaction through induction, which is explained by Faraday's Law: $\varepsilon = -N\frac{\Delta\Phi_B}{\Delta t}$ where ε is the EMF, N is the number of turns in the coil, $\Delta\Phi_b$ is the change in flux through the coil, and Δt is the change in time. Because the induced EMF depends on the rate of change in flux, any disturbance in the magnetic field caused by defects in the test metal can produce measurable differences, especially when looking at a range of frequencies [9].

Eddy Currents and Signal Interaction

When a conductive material is placed within the magnetic field generated by the transmission coil, circulating currents (eddy currents) are induced within the material due to a

changing magnetic flux [10]. This behavior is a fundamental principle behind electromagnetic non-destructive testing methods, where induced currents are used to evaluate material properties [9].

These currents generate opposing magnetic fields, consistent with Lenz's Law. Because of this, the presence of a conductive sample changes the overall magnetic field in the system and directly affects the signal detected by the pickup coil. Prior work shows that these interactions can alter both the amplitude and phase of the measured signal depending on the material and its condition [9].

This change in the magnetic field leads to a measurable difference in the received signal. When no material is present, the pickup coil measures only the field produced by the transmission coil. Once a metal sample is introduced, the opposing field generated by the eddy currents modifies the net field reaching the pickup coil. This results in a change in the peak-to-peak voltage (Vpp) recorded by the oscilloscope. This change depends on the electrical properties and physical condition of the material. In a uniform, unmodified sample, eddy currents are able to circulate more freely, producing a consistent response. When defects such as cuts, abrasions, or material removal are present, the flow of these currents is disturbed because the discontinuities interrupt the formation of continuous current paths [10]. This disruption leads to a different electromagnetic response.

It is important to note that this relationship is not strictly linear. The measured signal is influenced by several factors, including geometry, positioning, and electromagnetic coupling, which can lead to more complex variations in the response [9]. Because of this, changes in voltage are interpreted as indicators of differences in material condition rather than direct measurements of specific defect properties. The differences in electromagnetic responses between control and defective samples form the basis for detecting changes in material condition using the proposed system.

Skin Depth and Frequency Dependence

A factor that was very important when deciding the range of frequencies to use was the skin depth, which describes how the magnetic fields penetrate the test metal. The equation for skin depth is given by $\delta = \sqrt{\frac{2}{\omega\mu\sigma}}$, where ω is the angular frequency, μ is the magnetic permeability of the metal, and σ is the electrical conductivity. This relationship shows that as the frequency increases, skin depth decreases, meaning that the currents are confined to being closer to the surface of the metal. This behavior is known as the skin effect [11]. By testing at such a large range of frequency (4kHz to 310kHz) the system is able to probe different depths of the metal [9] [10]. Defects within the material disrupt the flow of eddy currents differently depending on where they are in reference to the surface of the metal, so the range of frequency is made to detect defects at many points [9].

Frequency Sweep and Defect Detection

By sweeping across a range of frequencies, the system captures how the material responds under varying conditions. Instead of relying on a single measurement, this produces a frequency profile that reflects differences in electromagnetic behavior between samples [12]. As a result, variations between control and defective samples can be observed across the frequency range, with the system expected to exhibit depth-dependent sensitivity due to the relationship between frequency and skin depth.

IV. Methodology

Construction of transmission / pickup coils

The transmission and pickup coils each consist of 100 turns of 30-gauge magnet wire around a 5-cm diameter 3-D printed spool. The length of the windings was selected to target an 8-ohm impedance at zero frequency to facilitate use with signal generators. Both the transmission coil and the pickup coil were wound as independent but identical coils around the same spool. This ensures that they maintain a constant relative position throughout testing, eliminating one source of spurious signal variation.

Development of the testing pipeline

To develop the testing system, we set up a processing pipeline in which a known input signal is generated, transmitted through a metal sample, and subsequently measured after being modified by the material. The pipeline consists of three main stages: signal generation, signal transmission, and signal acquisition.

Signal generation is handled via custom Python 3 script that utilizes the PyVISA library to communicate with a Siglent SDG 1032X function generator. The script iterates through a set of 15 frequencies logarithmically spaced between 4 kHz and 310 kHz. This range allows us to probe the sample from 1.08-mm deep to 0.118-mm deep and was chosen based on the capabilities of our hardware. At each step, it configures the parameters of a sinusoidal wave with the specified frequency, a constant amplitude of 2V, and zero offset. The script sends these parameters to the function generator for signal generation and waits for signal acquisition before proceeding to the next frequency (Figure 1.1).

Next, the function generator outputs the signal to the transmission coil (Figure 1.2,3). The metal to be tested is placed directly below the coil in contact with the spool. The signal generates an EMF in the pickup coil, which is measured by a Siglent SDS 1202X-E oscilloscope (Figure 1.4,5). We use the same Python script as before to communicate with the oscilloscope. To prevent clipping, the script dynamically adjusts the scale and the viewing window of the oscilloscope. This scaling is performed by first estimating the signal's peak-to-peak voltage and then adjusting the vertical division (V/div) so the waveform occupies approximately 75% of the display range, ensuring accurate measurement without clipping. The script then reads and saves the peak-to-peak voltage (Vpp) values to an array at an index that corresponds to the current frequency (Figure 1.6). Each sample was measured at a single representative region, and repeated measurements were not performed. Peak-to-peak voltage values were obtained using the oscilloscope's automatic measurement function via SCPI commands. The recorded values were reported with a resolution of 0.01 V; therefore, measurement uncertainty is estimated to be on the order of ±0.01 V, in addition to inherent instrument accuracy limitations. A schematic of the processing pipeline is shown in Figure 1, alongside a real-world image of the system for reference.

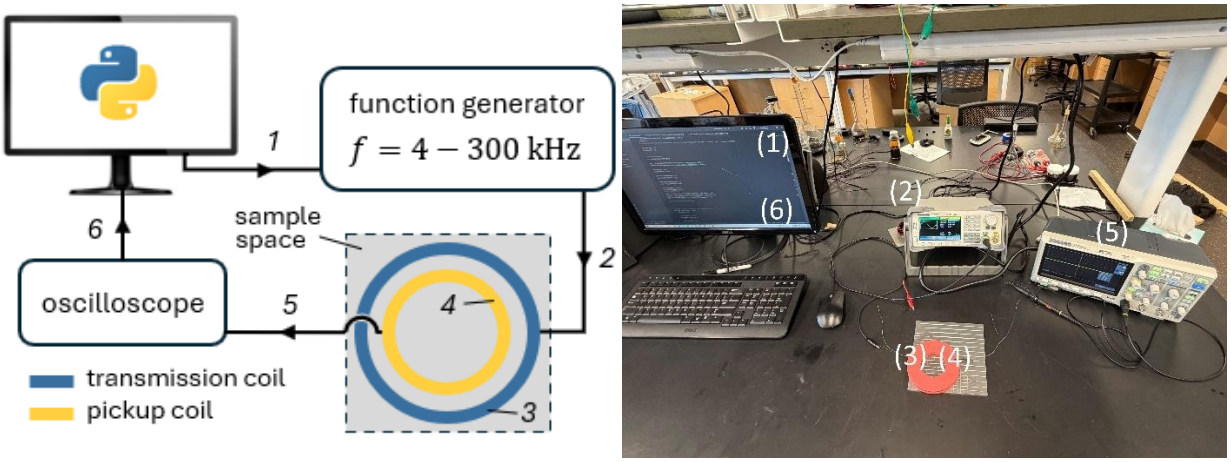


Figure 1. Schematic of processing pipeline with a real-world image provided to clarify component placement and sample positioning relative to the coils. (1) a Python script sends parameters to a function generator, (2) the function generator outputs the signal to the transmission coil, (3) the transmission coil passes the signal through the sample metal, (4) the sample metal modifies the signal and passes it to the pickup coil, (5) the pickup coil sends the modified signal to an oscilloscope, (6) the oscilloscope sends the final modified signal back to the Python script for analysis.

To enable faster testing, the system first sweeps through the whole set of fifteen frequencies with a control sample (unmodified aluminum) and saves the corresponding values to an array dedicated to control. It then repeats the process for the test sample (modified aluminum) and saves the corresponding values to an array dedicated to defects. These two arrays of peak-to-peak voltages are then graphed with each other versus frequency for data analysis (Figure 3). The total time required to complete a full frequency sweep was not fixed in the control script, but under the described testing conditions averaged approximately 45 seconds per sample, varying slightly due to signal settling time and instrument communication delays.

Structured test materials

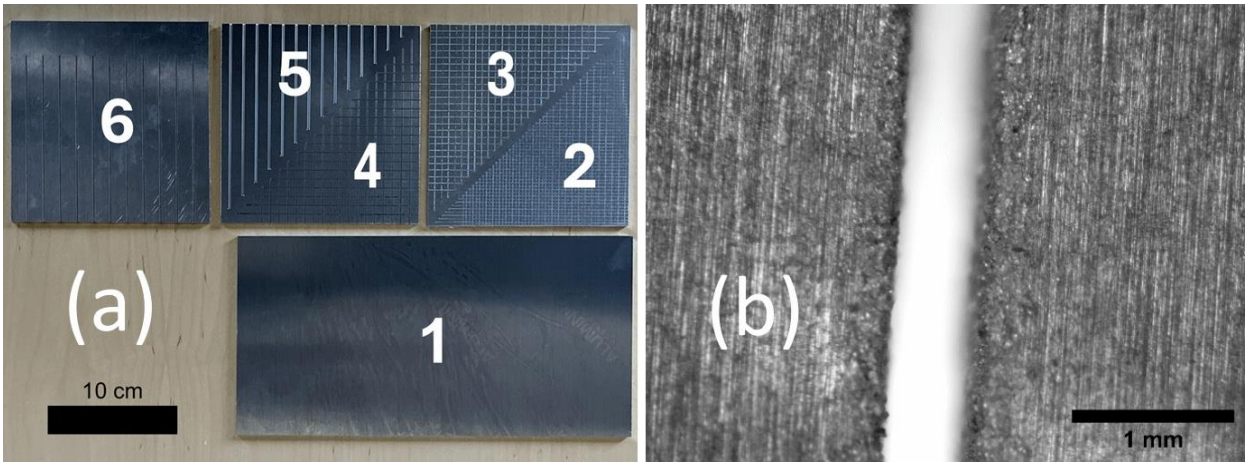


Figure 2. Control and test samples of aluminum. (a) Overview of all samples. A given 10-cm scale bar in the bottom left corner. The control sample (**1**) consists of an unmodified 3.2-mm thick aluminum sheet. Test samples **2-5** were modified with grooves cut with a 40º cutting tool on a CNC mill (HAAS, Super Mini Mill 2). Grid spacings and depths are: (**2**) 0.2-mm depth, 0.25-cm grid; (**3**) 0.5-mm depth, 0.50-cm grid, (**4**) 1.0-mm depth, 0.67-cm grid; (**5**) 1.5-mm depth, 1.00-cm grid. Test sample **6** was through-cut with a waterjet (Wardjet, A-Series) at 1.27-cm intervals with a 0.5-mm cut width, leaving a 2.54-cm boundary at the top. (b) Close-up of Sample 6 highlighting the through-cut defect thickness, with a 1-mm scale bar provided for clarity.

Aluminum was selected as the base material due to its non-ferrous nature and high electrical conductivity, which make it well-suited for electromagnetic induction-based testing. These properties allow for strong eddy current formation under an applied alternating magnetic field. In addition to its favorable electromagnetic behavior, aluminum is widely used in real-world structural applications where early defect detection is critical, particularly in aerospace systems where aircraft fuselages and components require consistent inspection for cracks and fatigue.

An unmodified aluminum sheet (Figure 2.1) was selected as the control sample. The aluminum measured 3.175-mm in thickness, with a length of 15.24-cm, and a width of 30.48-cm. It is important to note here that the areas of the samples do not play a critical role if they cover a diameter of more than 5-cm, since that area is sufficient to span the testing coil. This control sample establishes a baseline for aluminum under ideal conditions using the script described previously.

Afterwards, several purposeful and intentional defects were introduced to otherwise identical pieces of aluminum (Figure 2.2-6). Each piece was carefully manufactured to ensure a consistent and stable 'defect' throughout the pieces. The measurements of each modification are as follows:

Sample 2. 0.2-mm deep: 0.25-cm grid
Sample 3. 0.5-mm deep: 0.5-cm grid
Sample 4. 1.0-mm deep: 0.67-cm grid
Sample 5. 1.5-mm deep: 1-cm straight lines
Sample 6. Full depth, 1.27 cm between each cut; 0.5-mm cut width; 2.54-cm border on the top

V. Results and Discussion

To further analyze the experiment, additional testing conditions and data collection methods were incorporated. In addition to comparing control and modified metal samples, each sample was tested in two configurations, both above and below the coil to evaluate whether the relative placement of the material within the magnetic field affects the measured response. Through this process, it was determined that placing the material beneath the coil reduces interference and provides more consistent and accurate measurements.

With the finalized experimental setup, our modified samples (#2-#6) were tested against a control—Sample #1—across all frequencies. The results for the configuration with the material placed beneath the coil are shown in Figure 3.

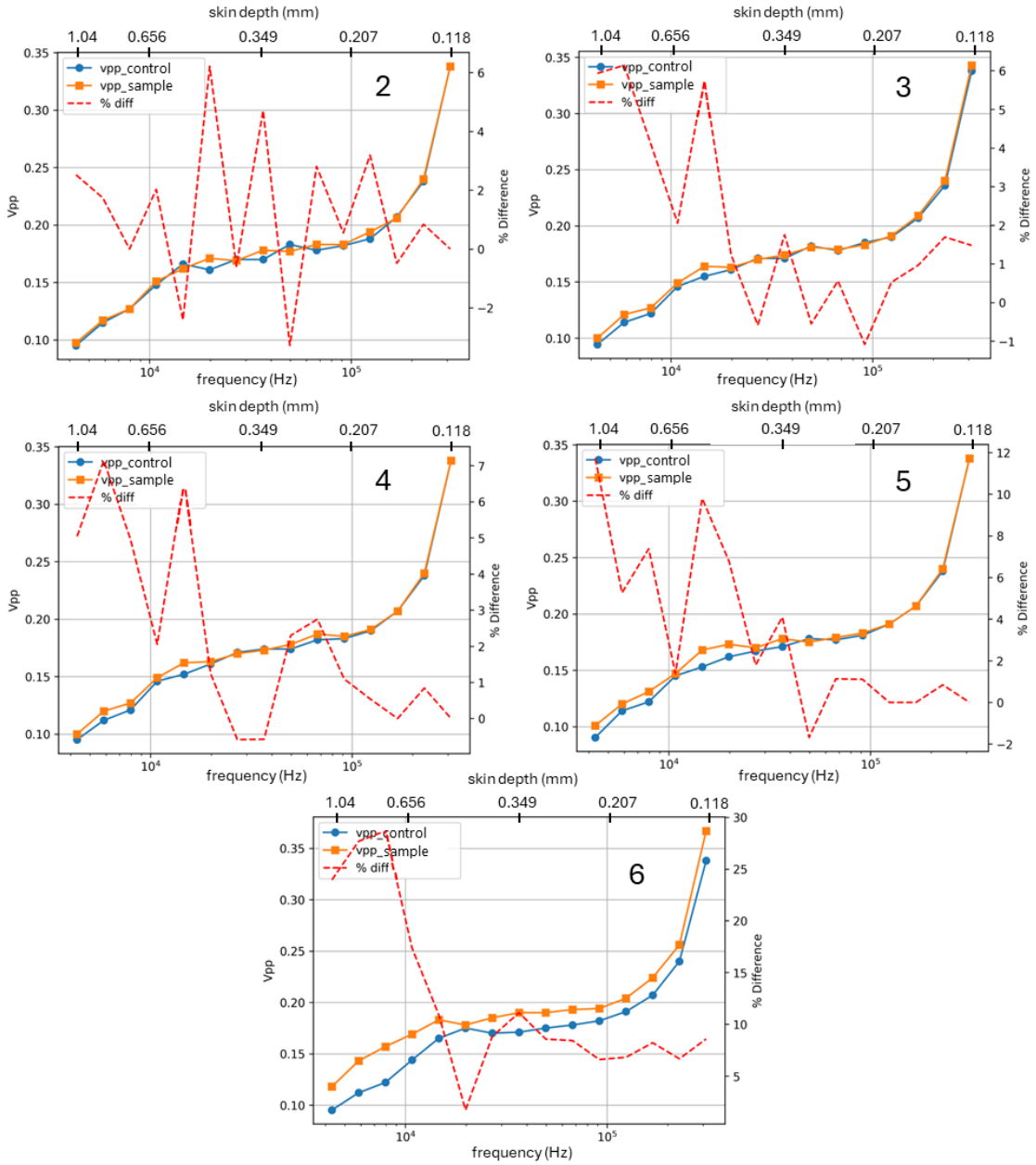


Figure 3: Peak-to-peak voltage vs. frequency for control and modified samples. Peak-to-peak voltage (Vpp) is plotted as a function of frequency for a control metal sample compared against multiple modified samples (Samples #2–#6). Each subplot represents a different modified sample, with the control measurement re-acquired for each comparison under the below-coil configuration. The percent difference (sample − control) is shown on the y-secondary axis. A secondary x-axis indicates the corresponding skin depth (mm), illustrating the inverse relationship between frequency and penetration depth.

For each configuration, peak-to-peak voltage (Vpp) values were recorded across the full set of frequencies, producing a frequency response curve for each material condition. These results allow for direct comparison between control and modified samples. In addition to the raw voltage measurements, further analysis was performed to better quantify differences between samples. This includes calculating the percent difference between control and modified signals at each frequency, computing the ratio between modified and control responses, and comparing the behavior of the signal at lower and higher frequencies. The overall response across all

frequencies was also evaluated by considering the total signal behavior over the entire frequency range.

These analysis methods provide a more detailed representation of how the electromagnetic response of the material varies with frequency and condition, allowing for clearer identification of differences between samples that may not be immediately visually apparent from raw voltage measurements alone.

While the frequency response curves enable direct comparison between control and modified samples, the majority of Samples (#2, #3, and #5) exhibit minimal deviation from the control signal across the tested frequency range, suggesting that the observed differences fall within experimental uncertainty. Sample #4 shows some deviation at lower frequencies; however, this behavior is inconsistent with expectations, as higher frequencies (corresponding to reduced skin depth) were anticipated to exhibit greater sensitivity to material defects.

In contrast, Sample #6 shows the most significant and consistent deviation from the control across the tested frequency range. This sample contained the most extreme defect, with material fully cut through the metal, representing a clear upper bound on defect sensitivity. The separation between the control and modified response is evident across frequencies, and the percent difference curve for Sample #6 (Figure 3.6) further quantifies this behavior. For Sample #2-5, the root-mean-square (RMS) difference was 2.7%, 3.1%, 3.3%, and 5.1%, respectively, while the RMS difference for sample 6 was significantly higher, at 14.6%. Although this system clearly detects large defects, the results do not demonstrate a consistent frequency-dependent increase in sensitivity, suggesting that additional investigation is needed to isolate the role of skin depth in the measured response.

These results confirm that the experimental system is capable of detecting substantial structural defects. While more moderate samples do not show clear deviations beyond experimental uncertainty, the response from Sample #6 validates the measurement approach, demonstrating its effectiveness when defect magnitude is sufficiently large. Overall, the results suggest that the current system has a detection threshold, where only sufficiently large defects produce measurable deviations in the electromagnetic response.

Despite the clear trends observed, a major source of uncertainty should be acknowledged. During testing, the oscilloscope was unable to accurately measure signals at higher frequencies, leading to incorrect peak readings. The error was traced to the use of the 30-gauge wire, which increased circuit resistance and introduced a cutoff effect in the measured signal amplitude. As a result, measurement artifacts were observed at higher frequencies, particularly above 500 kHz. To mitigate this issue, the maximum frequency was reduced to 310 kHz, which remains sufficient for detecting defects within the targeted depth range.

Most real-world structural systems incorporate ferromagnetic materials such as steel, which are expected to exhibit different electromagnetic behavior under this testing approach. Due to their higher magnetic permeability, these materials produce stronger magnetic field interactions and altered eddy current distributions compared to non-ferrous materials like aluminum. Additionally, effects such as hysteresis losses from repeated magnetization cycles further influence the measured response. The increased permeability also reduces skin depth, limiting signal penetration into the material. As a result, electromagnetic testing of ferromagnetic materials is more complex and may lead to differences in frequency response, suggesting that material-specific calibration is necessary when extending this method beyond non-ferrous systems. Additionally, real-world defects are often irregular and non-uniform in shape and distribution, and material composition, including alloys, may further influence the measured

response, which should be considered in future evaluations of the system.

VI. Conclusion

This work presented a variable-frequency AC testing system for detecting defects in metallic materials using electromagnetic induction. By sweeping across a wide range of frequencies and analyzing the resulting peak-to-peak (Vpp) response, the system captures changes in the electromagnetic behavior of metal samples caused by structural modifications. The results demonstrated that this approach is capable of identifying differences between control and defective samples, particularly when defects significantly disrupt current flow within the material.

While several modified samples (Samples #2, #3, and #5) exhibited minimal deviation from the control—suggesting that their effects fall within experimental uncertainty—Sample #6 displayed a clear separation across the range of frequencies. These findings indicate that the system is effective at detecting large defects and that a measurable detection threshold exists for this method. However, this system exhibited key limitations, including reduced measurement accuracy at higher frequencies, which constrained the usable frequency range. Despite these limitations, the results validate the feasibility of using frequency-domain electromagnetic analysis as a non-destructive, rapid, low-cost method for defect detection.

Future work should focus on expanding the analysis beyond peak-to-peak voltage comparisons by incorporating additional frequency-domain features. Prior work has demonstrated that pulsed multifrequency eddy current techniques combined with spectrogram-based features and machine learning methods, such as k-nearest neighbors, can enable estimation of defect characteristics including depth and length in conductive materials [12]. Integrating similar approaches could enhance the sensitivity of the proposed system and enable detection of smaller or more subtle defects.

To extend this approach to larger-scale structures, additional considerations beyond the scope of this work are required. Spatial coverage may be addressed through either larger-diameter coils, which increase the sensing area, or arrays of smaller coils for localized measurements. Operating frequency must also be adapted to the material, as lower frequencies provide greater penetration depth while higher frequencies are more sensitive to surface-level defects. As a result, the effective sensing depth of this technique is limited by both the operating frequency and the electromagnetic properties of the material, with highly conductive materials generally exhibiting shallower penetration depths. The current system is best suited for materials and geometries that fully span the coil area, while larger structures would require scanning or distributed sensing approaches for full coverage. In practical applications, scanning large structural components would likely require either mechanical translation of the sensing apparatus or parallelized measurements using multiple coils to reduce acquisition time.

Overall, this work establishes a strong foundation for scalable, non-destructive evaluation methods applicable to infrastructure monitoring and maintenance.

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